



Bases & Dimension

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Introduction



Price Problem



\$ 70'000



?



\$ 160'000



Introduction



\$ 70'000



?



\$ 160'000

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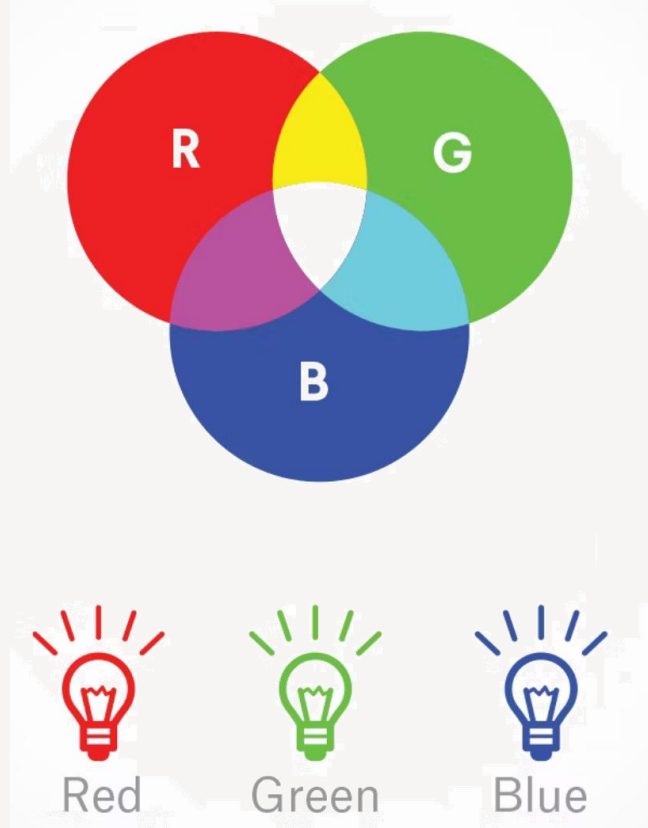
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Introduction



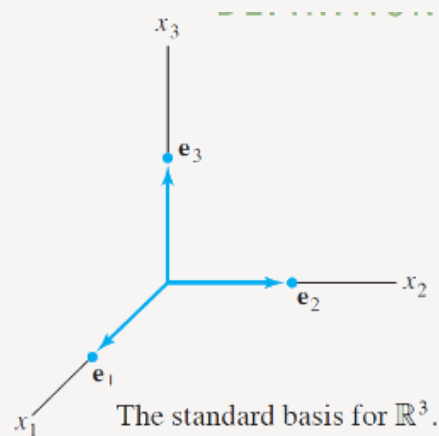
02

Basis



Basis

- A set of n linearly independent n -vectors is called a basis.
- A basis is the combination of span and independence: A set of vectors $\{v_1, \dots, v_n\}$ forms a basis for some subspace of $\mathbb{R}^n$ if it
 - (1) spans that subspace
 - (2) is an independent set of vectors.



Basis

Definition

Let H be a subspace of a vector space V . An indexed set of vectors $\mathcal{B} = \{b_1, \dots, b_n\}$ in V is a **basis** for H if

1. $\mathcal{B}$ is linearly independent set, and
2. The subspace spanned by $\mathcal{B}$ coincides with H ; that is,

$$H = \text{Span} \{b_1, \dots, b_n\}$$

Example

Which are unique?

- ☐ Express a vector in terms of any particular basis
- ☐ Bases for $\mathbb{R}^2$
- ☐ Bases with unit length for $\mathbb{R}^2$

Example Solution!

- **Express a vector in terms of any particular basis**

Unique: Once a basis is fixed, the coefficients (coordinates) are uniquely determined.

- 
- **Bases for $\mathbb{R}^2$**

Not unique: There are infinitely many possible bases, as any two linearly independent vectors can form one.

- **Bases with unit length for $\mathbb{R}^2$**

Not unique: There are infinitely many orthonormal bases (rotations of each other).



Vector Space of Polynomials

Be careful: A vector space can have many bases that look very different from each other!



Example (Basis)

- ☐ Standard bases for $P_n(\mathbb{R})$?
- ☐ Are $(1 - x), (1 + x), x^2$ basis for $P_2(\mathbb{R})$?

Example Solution!

- **Standard basis for $P_n(\mathbb{R})$:**

The standard basis is $\{1, x, x^2, \dots, x^n\}$.

Each polynomial in $P_n(\mathbb{R})$ can be uniquely written as a linear combination of these monomials.

- **Are $(1-x)$, $(1+x)$, x^2 a basis for $P_2(\mathbb{R})$?**

Yes. They are **linearly independent**.

$P_2(\mathbb{R})$ is **spanned** by these three vectors.

Therefore they form a valid basis for $P_2(\mathbb{R})$.

03

Dimension

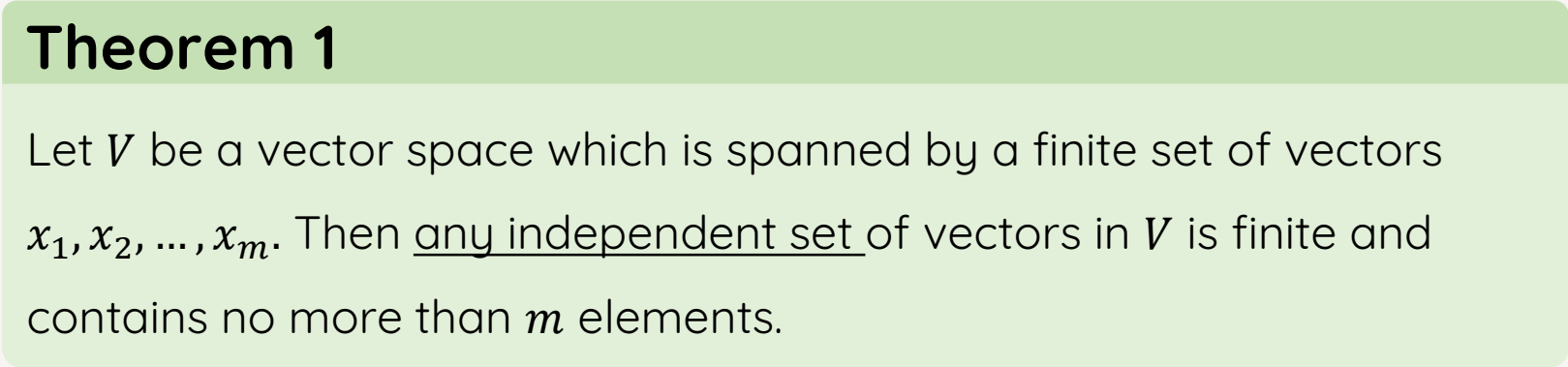


Dimensions

- The dimensionality of a vector is the number of coordinate axes in which that vector exists.
- If a vector space is spanned by a finite number of vectors, it is said to be **finite-dimensional**. Otherwise it is **infinite-dimensional**.
- The number of vectors in a basis for a finite-dimensional vector space V is called the dimension of V and denoted $\dim(V)$.

Bases and finite dimension

Theorem 1



Let V be a vector space which is spanned by a finite set of vectors $x_1, x_2, \dots, x_m$. Then any independent set of vectors in V is finite and contains no more than m elements.

Conclusion



Every basis of V is finite and contains no more than m elements.

Proof of Theorem 1

Because $\{x_1, \dots, x_m\}$ spans V , every vector in V can be written as a linear combination of these m vectors.

Now suppose we have a linearly independent set $\{y_1, \dots, y_n\}$ in V with $n > m$.

Each y_i can be expressed as

$$y_i = a_{i1}x_1 + \dots + a_{im}x_m$$

Consider a linear combination of the y_i 's that equals the zero vector:

$$0 = c_1y_1 + \dots + c_ny_n$$

Substitute each y_i in terms of the x_j 's:

$$(c_1a_{11} + \dots + c_na_{n1})x_1 + \dots + (c_1a_{1m} + \dots + c_na_{nm})x_m = 0$$

Since the x_j 's span V , this gives a homogeneous system of m equations in n unknowns $c_1, \dots, c_n$. If $n > m$, this system has a nontrivial solution (some $c_i \neq 0$), which means the y_i 's are linearly dependent. $\Rightarrow$ Contradiction!

Independent $\leq$ spanning

Conclusion

In a finite-dimensional space,

*the length of every
linearly independent list
of vectors*

$\leq$

*the length of every
spanning list of vectors*

Bases and finite dimension

Theorem 2

If V is a finite-dimensional vector space, then any two bases of V has the same (finite) number of elements.

Proof of Theorem 2

Let $B_1 = \{x_1, \dots, x_m\}$ and $B_2 = \{y_1, \dots, y_n\}$ be two bases of V .

Since B_1 spans V , any independent set (like B_2) can have **no more than** m elements $\Rightarrow n \leq m$.

Likewise, since B_2 spans V , any independent set (like B_1) can have **no more than** n elements $\Rightarrow m \leq n$.

Hence $m = n$.

Basis and finite dimension

The number of vectors in a basis for a finite-dimensional vector space V is called the dimension of V and denoted as $\dim(V)$.

Theorem 2



Theorem 3

Let V be a vector space with a basis B of size m . Then

- a) Any set of more than m vectors in V must be linearly dependent,
and
- b) Any set of fewer than m vectors cannot span V .

Proof of Theorem 3

- a) Since B is a basis, it spans V . By theorem 2, any linearly independent set in V can have at most m elements. Therefore, any set with more than m vectors must be linearly dependent.
- b) Suppose, for contradiction, that there exists a spanning set $\{y_1, \dots, y_k\}$ with $k < m$. From S , we can extract a basis (by removing dependent vectors). That basis would have at most k elements. But this contradicts the fact that every basis of V must have m elements.

Dimensions

Definition

A vector space V is called...

- a) **finite-dimensional** if it has a finite basis, and its **dimension**, denoted by $\dim(V)$, is the number of vectors in one of its bases.
- b) **infinite-dimensional** if it has no finite basis, and we say that $\dim(V) = \infty$.

Note

Dimension of subspace $\{\mathbf{0}\}$?

Dimensions

Example

Let's compute the dimension of some vector spaces that we've been working with.



Vector space	Basis	Dimension
$\mathcal{F}^n$ (n-tuples each elements from field $\mathcal{F}$)		
P^p (polynomials with max degree p)		
$M_{m,n}$ (matrices with m rows and n columns)		
P (all polynomials)		
F (all functions)		
C (all continues functions)		

Note!



04

Finite Dimensional Subspace



Basis of Subspace

Theorem 4

If W is a subspace of a finite-dimensional vector space V , every linearly independent subset of W is finite and is part of a (finite) basis for W .

Lemma

Let S be a linearly independent subset of a vector space V . Suppose u is a vector in V which is not in the subspace spanned by S . Then the set obtained by adjoining u to S is linearly independent.

Proof of Lemma

Assume

$$a_1 v_1 + \cdots + a_n v_n + cu = 0$$

If $c \neq 0$, we can solve for u : $u = -\frac{1}{c}(a_1 v_1 + \cdots + a_n v_n)$

This means u is a linear combination of vectors in S , i.e., $u \in \text{span}(S)$, which contradicts our assumption that $u \notin \text{span}(S)$.

Therefore, $c = 0$. The equation becomes $a_1 v_1 + a_2 v_2 + \cdots + a_n v_n = 0$.

Since S is linearly independent, this implies $a_1 = a_2 = \cdots = a_n = 0$.

Thus, all coefficients are zero, so $S \cup \{u\}$ is linearly independent.

Proof of Theorem 4

Let $\dim V = m$. Every linearly independent set in V has at most m elements. Hence any linearly independent subset of W also has size $\leq m$, so it is finite.

Therefore, the possible sizes of independent subsets of W lie in the finite set $\{0, 1, \dots, m\}$.

So largest possible size exists. Choose an independent subset $B \subseteq W$ that has this maximal size.

We claim B spans W . If not there exists a $w \notin \text{span}(B)$.

By lemma, $B \cup \{w\}$ must be linearly independent, contradicting maximality of B . Thus B is a finite basis for W .

Basis of Subspace

Corollary

A subspace is called a **proper subspace** if it's not the entire space, so $\mathbb{R}^2$ is the only subspace of $\mathbb{R}^2$ which is not a proper subspace

If W is a **proper subspace** of a finite-dimensional vector space V , then W is finite-dimensional and $\dim(W) < \dim(V)$

Corollary

In a finite-dimensional vector space V , every non-empty linearly independent set of vectors is part of basis.

Basis of sum of subspaces

Theorem 5

If W_1 and W_2 are finite-dimensional subspaces of a vector space V ,
the $W_1 + W_2$ is a finite-dimensional and

$$\dim(W_1) + \dim(W_2) = \dim(W_1 \cap W_2) + \dim(W_1 + W_2)$$

Proof of Theorem 5

Let $v_1, \dots, v_m$ be a basis of $W_1 \cap W_2$, so $\dim(W_1 \cap W_2) = m$.

Extend this basis to a basis of W_1 : $v_1, \dots, v_m, u_1, \dots, u_j$, so $\dim(W_1) = m + j$.

Similarly, extend $v_1, \dots, v_m$ to a basis of W_2 : $v_1, \dots, v_m, w_1, \dots, w_k$, so $\dim(W_2) = m + k$.

We will show that $v_1, \dots, v_m, u_1, \dots, u_j, w_1, \dots, w_k$ is a basis of $W_1 + W_2$.

1. Spanning:

Every vector in W_1 is a linear combination of v 's and u 's, and every vector in W_2 is a linear combination of v 's and w 's. Hence the span of all these vectors contains both W_1 and W_2 , so it equals $W_1 + W_2$.

Proof of Theorem 5 (cont.)

2. Linear Independence:

Suppose $a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j + c_1w_1 + \cdots + c_kw_k = 0$.

Move the w -terms to one side:

$$-(a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j) = c_1w_1 + \cdots + c_kw_k.$$

The right-hand side is in W_1 , and the left-hand side is in W_2 ; hence this vector is in $W_1 \cap W_2$, which is spanned by $v_1, \dots, v_m$. Thus

$$d_1v_1 + \cdots + d_mv_m = c_1w_1 + \cdots + c_kw_k \text{ for some } d_i\text{'s.}$$

Since $v_1, \dots, v_m, u_1, \dots, u_j, w_1, \dots, w_k$ are linearly independent all c_i 's (and d_i 's) are 0, so we have: $a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j = 0$ which implies all a_i 's and b_i 's 0 since these vectors are basis of W_1 . Hence all vectors are linearly independent.

Basis of sum of subspaces

Theorem 6

If W_1 , W_2 and W_3 are finite-dimensional subspaces of a vector space V , then can we have the following relation?

$$\begin{aligned} & \dim(W_1 + W_2 + W_3) \\ &= \dim(W_1) + \dim(W_2) + \dim(W_3) - \dim(W_1 \cap W_2) \\ & \quad - \dim(W_2 \cap W_3) - \dim(W_1 \cap W_3) + \dim(W_1 \cap W_2 \cap W_3) \end{aligned}$$

Counterexample: $W_1 = \text{span}\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right\}$, $W_2 = \text{span}\left\{\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}$, $W_3 = \text{span}\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$

Basis of sum of subspaces

Theorem 6

If W_1 , W_2 and W_3 are finite-dimensional subspaces of a vector space V , then:

$$\begin{aligned} & \dim(W_1 + W_2 + W_3) \\ & \leq \dim(W_1) + \dim(W_2) + \dim(W_3) - \dim(W_1 \cap W_2) \\ & \quad - \dim(W_2 \cap W_3) - \dim(W_1 \cap W_3) + \dim(W_1 \cap W_2 \cap W_3) \end{aligned}$$

Dimensionality and Properties of Bases

Note

Let V be a finite dimensional vector space over field F . Below are some properties of bases:

1. Any linearly independent list can be extended to a basis (a maximal linearly independent list is spanning).
2. Any spanning list contains a basis (a minimal spanning list is linearly independent).
3. Any linearly independent list of length $\dim V$ is a basis.
4. Any spanning list of length $\dim V$ is a basis.

We will learn about change of basis later.

05

Coordinates



Ordered basis

Definition

If V is a finite-dimensional vector space, an **ordered basis** for V is a finite **sequence** of vectors which is linearly independent and spans V .

Be careful: The order in which the basis vectors appear in B affects the order of the entries in the coordinate vector. This is kind of janky (technically, sets don't care about order), but everyone just sort of accepts it.

Coordinate Systems

- The main reason for selecting a basis for a subspace H ; instead of merely a spanning set, is that **each vector in H can be written in only one way as a linear combination of the basis vectors.**

Note

Suppose the set $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ is a basis for a subspace H . For each x in H , the **coordinates of x relative to the basis $\mathcal{B}$** are the weights $c_1, \dots, c_p$ such that $x = c_1 \mathbf{b}_1 + \dots + c_p \mathbf{b}_p$, and the vector in $\mathbb{R}^p$

$$[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix}$$

is called the **coordinate vector of x (relative to $\mathcal{B}$)** or the $\mathcal{B}$ -coordinate vector of x .

Coordinate Systems

Example

Coordinate vector of $p(x) = 4 - x + 3x^2$ respect to basis $\{1, x, x^2\}$



Example Solution!

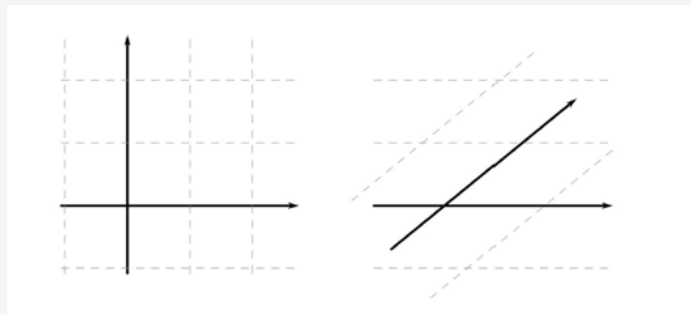
$$p(x) = 4(1) - 1(x) + 3(x^2)$$

$$[p(x)]_B = \begin{bmatrix} 4 \\ -1 \\ 3 \end{bmatrix}$$



Coordinate axes

- ❑ The familiar Cartesian plane (left) has orthogonal coordinate axes. However, axes in linear algebra are not constrained to be orthogonal (right), and non-orthogonal axes can be advantageous.



Resources

- ❑ Page 97 LINEAR ALGEBRA: Theory, Intuition, Code
- ❑ Page 213: David Cherney,
- ❑ Page 54: Linear Algebra and Optimization for Machine Learning

